

## 0-1 Sequences

CPU TIME LIMIT

1 second

MEMORY LIMIT

1024 MB

DIFFICULTY

medium (3.4)

You are given a sequence, in the form of a string with characters '0', '1', and '?' only. Suppose there are  $k$  '?'s. Then there are  $2^k$  ways to replace each '?' by a '0' or a '1', giving  $2^k$  different 0-1 sequences (0-1 sequences are sequences with only zeroes and ones).

For each 0-1 sequence, define its number of inversions as the minimum number of adjacent swaps required to sort the sequence in non-decreasing order. In this problem, the sequence is sorted in non-decreasing order precisely when all the zeroes occur before all the ones. For example, the sequence 11010 has 5 inversions. We can sort it by the following moves: 11010  $\rightarrow$  11001  $\rightarrow$  10101  $\rightarrow$  01101  $\rightarrow$  01011  $\rightarrow$  00111.

Find the sum of the number of inversions of the  $2^k$  sequences, modulo 1 000 000 007 ( $10^9 + 7$ ).

### Input

The first and only line of input contains the input string, consisting of characters '0', '1', and '?' only, and the input string has between 1 to 500 000 characters, inclusive.

### Output

Output an integer indicating the aforementioned number of inversions modulo 1 000 000 007.



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### Sample Input 1

?0?

### Sample Output 1

3

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